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Network Analysis of the Information Consumption-Production Dichotomy in Mastodon User Behaviors

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What are Decentralized Online Social Networks?

- Centralized social platforms typically see sole players as owners, e.g., companies:
  - Lack of "owner scalability"
  - Potential privacy and policies concerns
- DOSNs as an established alternative to the popular centralized counterparts, to bring back the user at the center of the stage

How is decentralization achieved?

Open-Source Software

Allows anyone to create a new server, or *instance*, thus favoring the emergence of communities guided by spontaneous interest towards certain topics

Communication Protocols

Enable seamless communication between (users registered on) different instances, even if pertaining to heterogeneous services

Development of the ***Fediverse***:
the *federated* universe of decentralized instances

Our Goal

Understand how the seamless interaction between users of different instances impacts on the users' contribution to the community life, and their resulting role(s), through the information consumption-production dichotomy

Research Questions

- Do the consumers, resp. producers, identified in one instance exhibit the same behavior consistently while interacting with other instances?
- Are there users who behave as consumers in one instance and simultaneously as producers in other instances, or vice versa?

Dataset

We used the most complete and up-to-date network dataset of Mastodon:

- More than 1.3M unique users and about 17M unique links between users
- Coverage of ~78% of the Mastodon user base to date

Lucio La Cava, Sergio Greco, and Andrea Tagarelli. **Understanding the growth of the Fediverse through the lens of Mastodon.** Appl. Netw. Sci., 6(1):64, 2021

Lucio La Cava and Andrea Tagarelli. **Information Consumption and Boundary Spanning in Decentralized Online Social Networks: the case of Mastodon Users.** CoRR, abs/2203.15752, 2022

Network Modeling - Mastodon User Network

$\mathcal{U}$ set of users, $\mathcal{I}$ set of instances

$\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$ directed graph

$\mathcal{V}$ contains pairs (u, i) s.t. $u \in \mathcal{U} \wedge i \in \mathcal{I}$

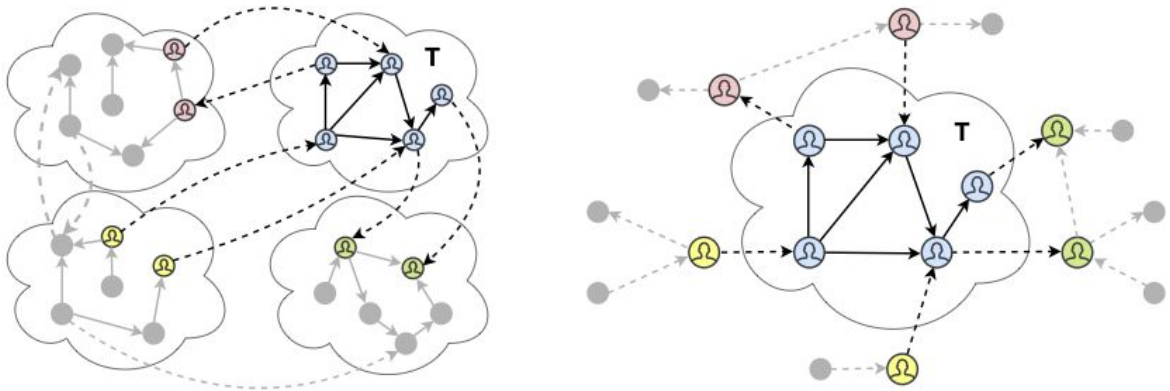
$\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ s.t. any $(x, y) \in \mathcal{E}$ with $x = (u, i) \wedge y = (v, j)$
means that user v in instance j **receives information produced by** user u in instance i (u may coincide with v provided that $i \neq j$)

Network Modeling - Extended ego-network

Given a specific instance $i \in \mathcal{I}$

$\mathcal{G}_i = \langle \mathcal{V}_i, \mathcal{E}_i \rangle$ directed subgraph induced from $\mathcal{G}$ s.t. $\mathcal{V}_i \subseteq \mathcal{V}$ and

$$\mathcal{E}_i = \{(x, y) |\ x = (u, j),\ y = (v, k) \wedge (j = i \vee k = i)\} \subseteq \mathcal{V}_i \times \mathcal{V}_i$$



Methodology and Assessment Criteria

1. Locate the top-25 Mastodon instances by relevance, group them into contexts, i.e., top-5/10/25, and for each instance in a given context, induce the ego-network
2. Leverage the LurkerRank, a content-agnostic eigenvector-centrality-based node ranking method to shape lurking behaviors within ego-networks

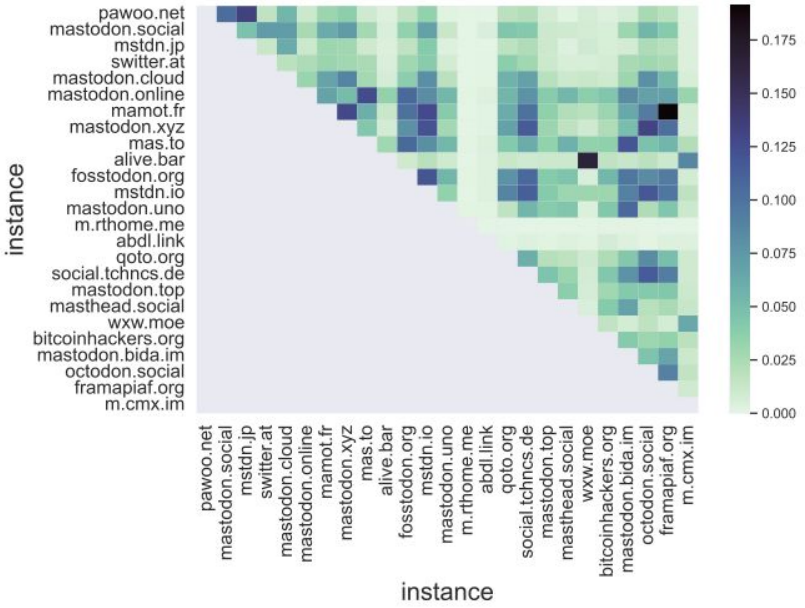
Tagarelli, A.; and Interdonato, R. 2014. **Lurking in social networks: topology-based analysis and ranking methods.** Soc. Netw. Anal. Min., 4(1): 230.

Methodology and Assessment Criteria

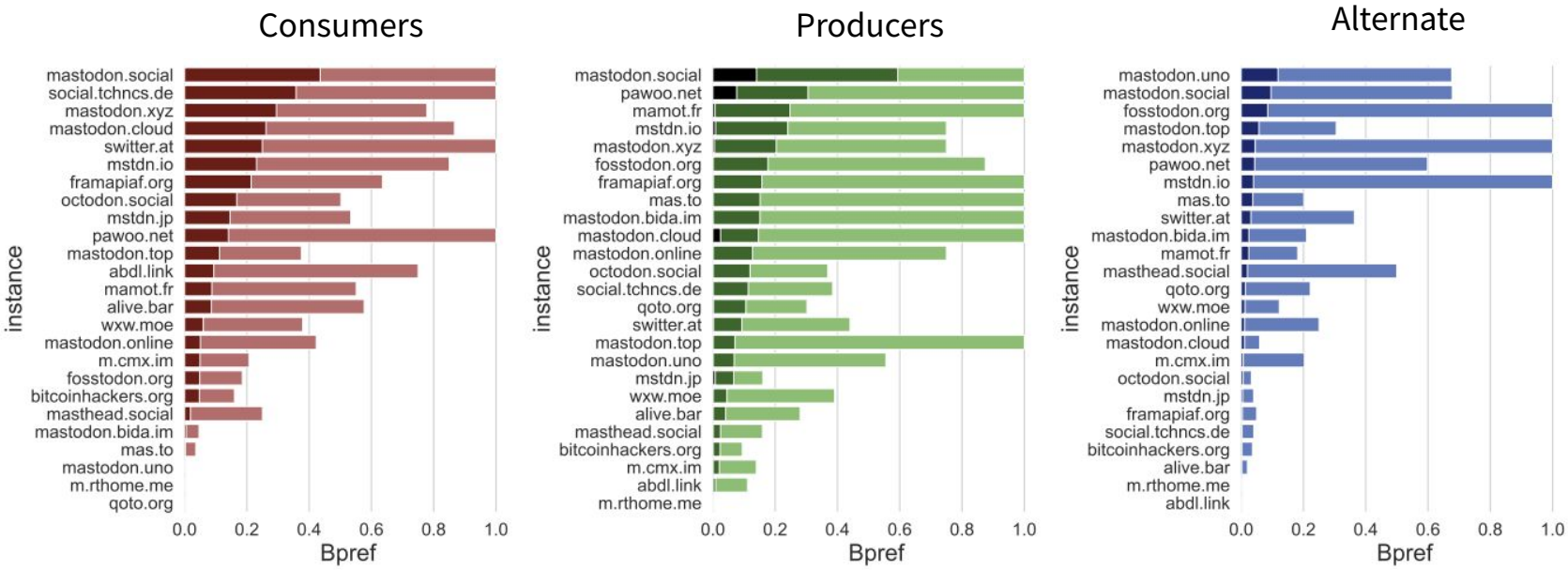
3. Use the Jaccard similarity index to study behavioral consistency, resp. alternation, within contexts
4. Evaluate the Binary preference conditioning scores of each instance ego-network w.r.t. its context, to gain deeper insights into the behavioral consistency/alternation phenomena

Results - Overview

Behavioral consistency/alternation in a small number of largest Mastodon instances, limited to pairwise instance comparisons



Results - Behavioral Consistency vs. Alternation



Thanks!

Any questions?

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